

E-Commerce Sales & Customer Analytics Dashboard

Power BI Project Documentation

Tools: Excel • Power Query • Power BI • DAX

Project focus: Sales performance, profitability, product performance, customer behavior, geography, and discount analysis.

1. Project Overview

This project presents an end-to-end e-commerce analytics solution built in Microsoft Power BI. Raw Excel data was cleaned and transformed using Power Query, modeled through relationships, and analyzed using DAX measures. The final interactive dashboard provides business-focused views of sales, profitability, products, customers, geography, monthly trends, and discounts.

Workflow: Raw Excel Data → Data Cleaning → Data Modeling → DAX Measures → Dashboard → Business Insights

2. Business Problem

An e-commerce business needs a consolidated view of performance to understand revenue, profitability, customer contribution, product performance, regional performance, and discount behavior.

- How much sales and profit are being generated?
- Which products and categories contribute the most revenue?
- Which customers generate the highest sales?
- Which states and cities perform best?
- How does sales performance change month by month?
- How do discounts relate to profitability?
- What data-quality issues need attention?

3. Project Objectives

- Analyze overall sales and profitability.
- Track key business KPIs.
- Identify top-performing products and categories.
- Analyze customer purchasing performance.
- Understand geographical sales distribution.
- Evaluate monthly sales trends.
- Analyze discount patterns and profitability.
- Build an interactive, business-focused Power BI dashboard.
- Demonstrate an end-to-end data analytics workflow.

4. Dataset & Tables

Table	Purpose	Key Columns
Customers	Customer information	CustomerID, CustomerName, Gender, City, State

Table	Purpose	Key Columns
Products	Product and pricing information	ProductID, ProductName, Category, UnitPrice, CostPrice
Orders	Order-level information	OrderID, CustomerID, OrderDate, SalesChannel, Status
Order_Details	Order-line transactions	OrderID, ProductID, Quantity, UnitPrice, Discount
Date	Time analysis	Date, Year, MonthNo, Month, Quarter

5. Data Cleaning & Transformation

Customers

- Removed the duplicate customer record.
- Standardized data types and verified identifiers.

Products

- Standardized inconsistent category spelling to Stationery.
- Filled the missing USB Hub UnitPrice using supporting transaction evidence.
- Created CostPrice using $\text{UnitPrice} \times 70\%$ as an analytical cost assumption.

Orders

- Converted OrderDate from text to Date using the appropriate locale.
- Standardized online to Online.
- Retained an unknown Order Status where the correct value could not be established.

Order Details

- Standardized numeric data types.
- Changed the missing Discount to 0 where appropriate.
- Retained the missing Quantity rather than inventing a value.

Validation

- Checked all five tables for Power Query errors: 0 errors.

6. Data Model & Relationships

The report uses a relational, star-like analytical model. Customers, Products, and Date support analysis around the transactional Orders and Order_Details tables.

- Customers[CustomerID] 1 → * Orders[CustomerID]
- Orders[OrderID] 1 → * Order_Details[OrderID]
- Products[ProductID] 1 → * Order_Details[ProductID]
- Date[Date] 1 → * Orders[OrderDate]

Relationships were configured as one-to-many with single-direction filtering and active relationships.

7. DAX Measures

Total Sales

```
Total Sales =  
SUMX(  
    Order_Details,  
    Order_Details[Quantity] * Order_Details[UnitPrice] * (1 - Order_Details[Discount])  
)
```

Total Orders

```
Total Orders = DISTINCTCOUNT(Orders[OrderID])
```

Total Customers

```
Total Customers = DISTINCTCOUNT(Customers[CustomerID])
```

Total Quantity

```
Total Quantity = SUM(Order_Details[Quantity])
```

Average Order Value

```
Average Order Value =  
DIVIDE([Total Sales], [Total Orders])
```

Total Profit

```
Total Profit =  
SUMX(  
    Order_Details,  
    Order_Details[Quantity] * (Order_Details[UnitPrice] * (1 - Order_Details[Discount]) -  
    RELATED(Products[CostPrice]))  
)
```

Profit Margin %

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

Average Discount %

```
Average Discount % = AVERAGE(Order_Details[Discount])
```

8. Dashboard Pages

Executive Sales Overview

High-level performance view with KPI cards, Monthly Sales Trend, Sales by Category, and Sales by State.

- Total Sales: ■341.07K
- Total Profit: ■89.53K
- Total Orders: 20
- Total Customers: 12
- Total Quantity: 49
- Average Order Value: ■17.05K
- Profit Margin: 26.25%

Product & Profitability Analysis

Product and profitability analysis.

- Top Products by Sales
- Category Sales vs Profit
- Discount vs Profitability — Scatter Chart

Customer Analysis

Customer contribution and geographical analysis.

- Top Customers by Sales
- Customer Performance Matrix
- Sales by City
- Interactive slicers: City, Gender, Sales Channel

9. Key Business Insights

Overall Performance

The business generated approximately ■341.07K in sales and ■89.53K in profit, producing a 26.25% profit margin.

Monthly Sales Trend

January was the strongest month. Sales declined significantly in February, partially recovered in March, and declined again in April, indicating meaningful monthly volatility.

Category Performance

Electronics is the strongest revenue-generating category, followed by Furniture. Accessories and Stationery contribute comparatively lower sales.

Product Performance

Laptop is the leading product by sales, making availability and complementary-product cross-selling relevant opportunities.

Geographical Performance

Tamil Nadu is the strongest-performing state, with Chennai a major city contributor.

Customer Performance

A small group of high-value customers contributes a significant share of sales, creating opportunities for retention and personalized engagement.

Discount & Profitability

Higher discounts do not automatically produce higher profitability, so discount decisions should be evaluated at the product level.

10. Business Recommendations

- Maintain strong inventory availability for top-selling products, particularly laptops and other electronics.
- Use loyalty and personalized offers to retain high-value customers.
- Investigate the causes of monthly sales fluctuations.
- Evaluate discount strategies to balance sales growth and profit protection.
- Explore growth opportunities in lower-performing categories.
- Strengthen regional marketing based on state and city performance.
- Improve data-entry validation to reduce duplicate, inconsistent, and missing data.

11. Data Quality & Analytical Judgment

An important aspect of the project was distinguishing between values that could be reasonably inferred and values that should remain unknown. The missing USB Hub UnitPrice was supported by transaction evidence and could be filled. In contrast, a missing transaction Quantity could not be reliably determined and was retained as blank. This avoids introducing fabricated information into business calculations.

12. Tools & Technologies

- Microsoft Excel — source data
- Power Query — data cleaning and transformation
- Microsoft Power BI — data modeling, visualization, and dashboard development
- DAX — business calculations and KPIs

13. Project Outcome

The completed solution transforms raw e-commerce data into an interactive analytical dashboard. It demonstrates practical skills in data preparation, relational modeling, DAX, visualization, business interpretation, and data-quality handling. The final report is structured for both stakeholder analysis and portfolio presentation.

Author

Reshma Krishnapillai

Aspiring Data Analyst | Excel | SQL | Power BI | Python